

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Even Semester)

Innovative Practices Description

UNIT II Cellular Architecture

Degree, Semester& Branch: VI Semester B.E. ECE B
Course Code & Title: EC8652 Wireless Communication
Name of the Faculty member: Mrs.G.Gnana Priya
Name of the Topic: Multiple Access Techniques
Name of the Innovative Practice: Think Pair Share
Date & Duration: 02.01.2020 & 20 minutes

Description of Think Pair Share:

Think Pair Share is a co-operative discussion strategy that was developed by Frank Lyman and some of his colleagues in the early 1980's. It has three phases. During phase 1(Think) the instructor poses a question and allows the students time to ponder their response. During phase 2 (Pair) the student discuss their thoughts with their designated partner or tablemate. . During phase 3 (Share) the teacher will call upon the groups to have them tell the class what they discussed and think about that topic. This activity increases the collaborative learning skills of the students.

Goals(Learning Outcome):

The students will be able to analyze and compare the various multiple access techniques.

Use of Appropriate methods:

Justification for choosing this topic using Think Pair Share Activity:

Since multiple access technique is an important concept in exam point of view, this activity will help the students to know about the comparison of various multiple access techniques. So I have chosen this topic to conduct Think-pair-Share activity.

Effective Presentation:

Details of Implementation:

- **Think(T)** : During the first phase (T), I asked the students to think (individually) about the comparison among various multiple access techniques
- **Pair(P)** : During the second phase (P), the students were asked to pair with their neighbours to discuss about what they think about the posed question
- **Share(S)** : During the third phase (S), the students shared their thinking with their tablemate.

After this Sharing phase, I randomly called five groups to share what they discussed about the topic with the whole class.

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

Since I conducted this activity, the collaborative learning skills of the students got increased. They actively participated in this activity. The success of the activity was felt when asked questions in the classroom.

Sample images:



Reflective Critique:

- **Benefits of conducting this activity:**
 - ✓ This activity limits off-task behaviour and talking.
 - ✓ It increases the co-operative learning of the students.

- **Challenges in conducting the activity:**

- Noise level of the class room goes up and a transition was needed to get back on the topic.
- It was hard to keep every student on task.
- It took a longer time than other strategies.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	2	3	2	1	1			2		2		2

CO2: The students will be able to apply the various cellular concepts like frequency reuse, channel assignments, handoff strategies etc., in mobile communication.

References:

1. Rappaport, T.S., "Wireless communications", Second Edition, Pearson Education, 2010
2. https://nptel.ac.in/content/storage2/nptel_data3/html/mhrd/ict/text/117102062/lec31.pdf

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Innovative Teaching Method Execution

UNIT II Cellular Architecture

Degree, Semester& Branch: VI Semester B.E.ECE B

Course Code & Title: EC8652 Wireless Communication

Name of the Faculty member: Mrs.G.Gnana Priya

Name of the Topic: Cellular Architecture overview

Name of the Innovative Practice: Mind Map

Date & Duration: 09.01.2020 & 15 Minutes

Description of Mind Map Activity:

Mind map is a tool that incorporates graphics, words, numbers, and color that represent and reflect a specific lesson or concept. It is a graphical way to represent ideas and concepts. Mind map is a visual thinking tool that helps structuring information, helping the students to better analyze, comprehend, synthesize, recall and generate new ideas. It is an easy way to brainstorm thoughts organically without worrying about order and structure. It allows the student to visually structure the ideas which is used to help with analysis and recall.

Goals (Learning Outcomes):

- ❖ The students will be able to elaborate the cellular architecture concepts.

Use of appropriate methods:

Justification for choosing the topic using mind map activity:

The cellular architecture consists of various concepts and terms which is the very basic for wireless communication. So to make the students to recall all the concepts, I have chosen this topic for conducting mind map activity since it is an easy way to make all the students to participate in this activity.

Effective presentation

Details of the Implementation:

I called the students one by one to come to the stage and write the term or concept related to this cellular architecture. So it was easy for them to recollect the points learnt in the classroom and they were able to view all the terms in a single activity. It was very useful for the students and they were also be able to reproduce the concepts in the exams.

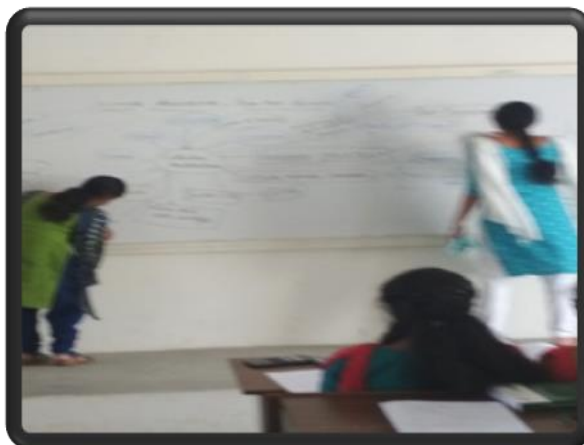
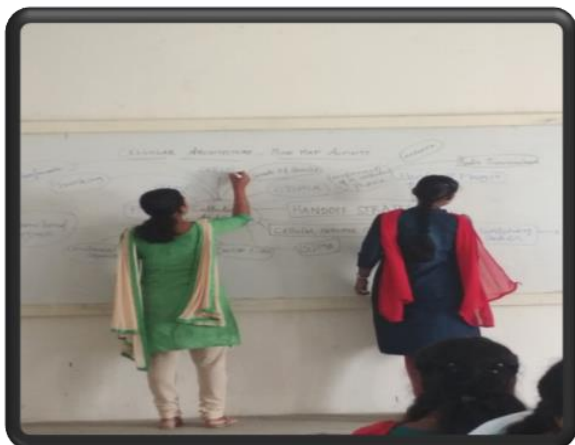
This mind map activity gives the graphical representation of words, concepts or pictures linked to and arranged around a given topic using a non-linear graphical layout which helps the students to build framework around a central concept.

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

By conducting this mind map activity, the students were able to recall the topics they had learnt in the classroom and they were also able to reproduce it in the exam. The assessment of effectiveness of the activity was felt when told most of the points.

Samples images of the activity:



Reflective Critique:

- **Benefits of conducting this activity:**

- ✓ It facilitates better understanding of relationships and connections between ideas and concepts of cellular architecture
- ✓ All the students actively participated in this activity and they asked me to conduct more such activities like this.

- ✓ This mind map activity helped the students to note down the most important terms or information using key words and then make them connection between the facts and ideas visually keeping all of the thoughts together on the black board.

- **Challenges:**

Since all the students eagerly participated in this activity, I felt no challenge in conducting this activity.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	2	1	1			2		2		2

CO1: The students will be able elaborate the characteristics of a wireless channel and evolve the system design specifications.

References:

1. Rappaport, T.S., "Wireless communications", Second Edition, Pearson Education, 2010
2. <https://www.youtube.com/watch?v=kxLcwIMYmr0>
3. <https://www.youtube.com/watch?v=cJQZvxfDFug>

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Innovative Practices Description

UNIT I WIRELESS CHANNELS

Degree, Semester& Branch: VI Semester B.E.ECE B

Course Code & Title: EC8652 Wireless Communication

Name of the Faculty member: Mrs.G.Gnana Priya

Name of the Topic: Propagation mechanism

Name of the Innovative Practice: Sketch Noting

Date & Duration: 11.01.2020 & 10 minutes

Description of Sketch Noting:

Sketchnotes are rich visual notes created from a mix of handwriting, drawings, hand-drawn typography, shapes, and visual elements like arrows, boxes, and lines. It is a strategy that engages every student so that checking the understanding of the whole class can be done efficiently.

Goals(Learning Outcome):

The students will be able to elaborate the propagation mechanism .

Use of Appropriate methods:

Justification for choosing the topic using Sketch Noting Activity:

In general the drawing and writing habit of the students gets reduced while they study a topic or concept. To make them visualize the concept by making them to write/draw the concept in the classroom itself increases their understanding level in a better way. So I have chosen this topic propagation mechanism to conduct sketch noting activity.

Effective Presentation:

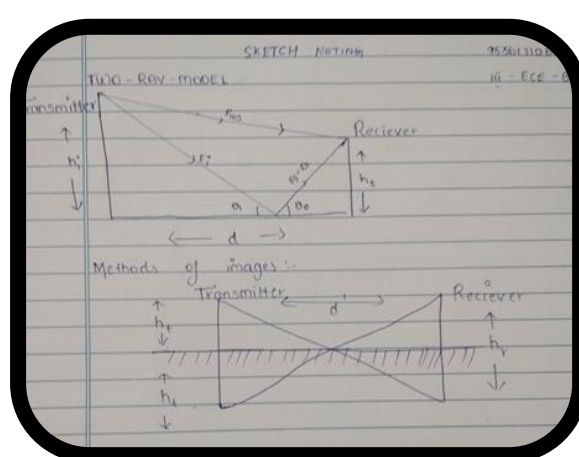
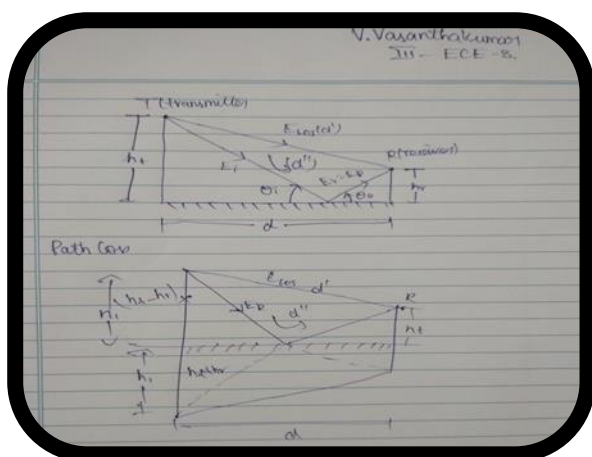
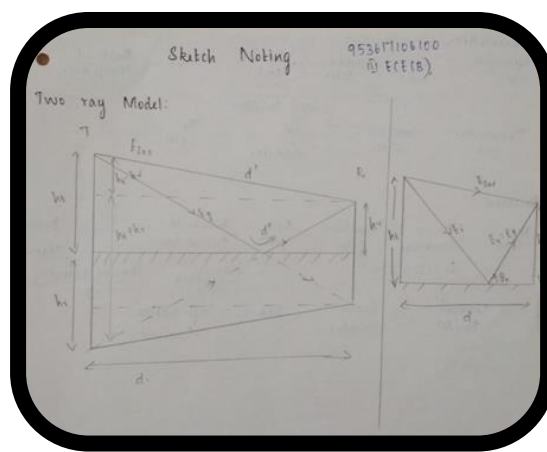
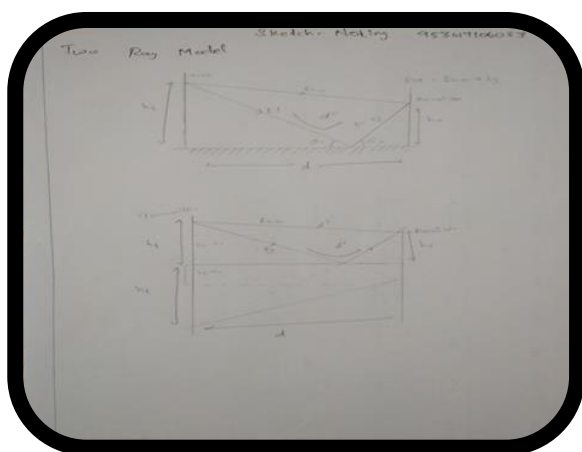
Details of Implementation:

After the topic was taught in the class, I asked the students to write/draw the propagation mechanism in a piece of paper.

Reflective Critique:

- **Benefits of conducting this activity:**
 - ✓ The active learning skills of the students got increased.
 - ✓ The students got a clear idea about the concept in the classroom itself.

Sample Images:



Challenges faced in conducting this activity:

Few students forgot the terms used in the two ray model (method of images) and they referred the notes to draw it. So it took few more minutes than what I planned to conduct this activity.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12

CO1: The students will be able to elaborate the characteristics of a wireless channel and evolve the system design specifications.

References:

1. Rappaport, T.S., "Wireless communications", Second Edition, Pearson Education, 2010.
2. Andreas.F. Molisch, -Wireless Communications, John Wiley – India, 2006

